

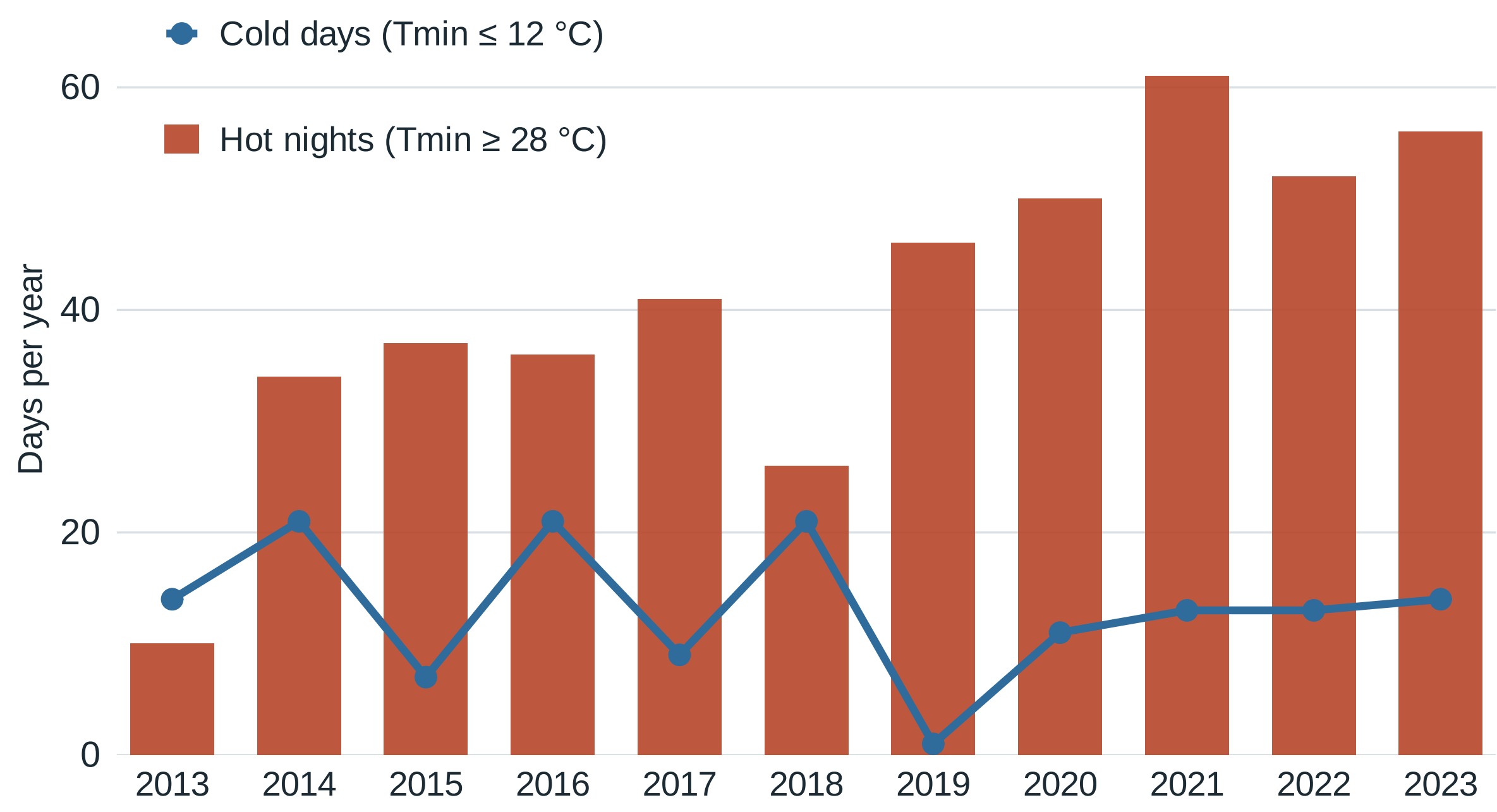
Heat, cold and first hospitalisation after cardiovascular diagnosis in Hong Kong

An exploratory monthly study, 2013–2023

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INTRODUCTION

Hong Kong’s hot nights became more frequent over the past decade: at the Hong Kong Observatory they rose from 10 days in 2013 to a peak of 61 in 2021^[1], while winter cold days have not disappeared. For the growing number of people living with type 2 diabetes or hypertension — a group at high cardiovascular risk — hospital burden^[2,3] complements the better-studied mortality burden^[4,5]. This study examines the monthly association between temperature, extreme-day burdens and first hospitalisation after cardiovascular diagnosis in that group.



Official Hong Kong Observatory extremes at Headquarters. Hot nights rose about fivefold over the decade; cold days recurred every winter — the exposure setting for this study.

RESULTS

Model 1: twelve thermal fits, specified after the outcome series was available — two residual signals, with unequal robustness.

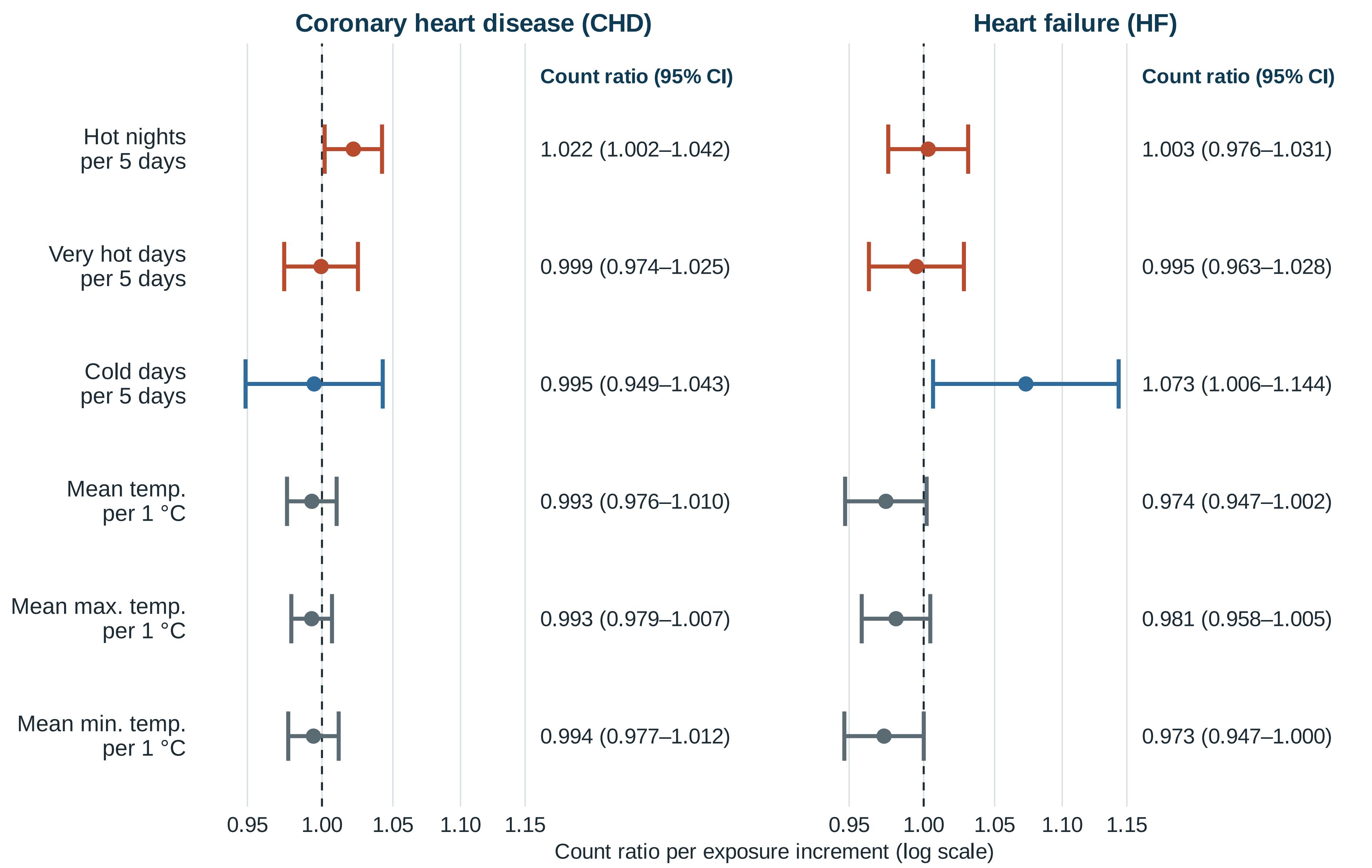
1.022

CHD series × hot nights, per 5 days
Full window (NW6): 1.002–1.042 $p = 0.032$ $q = 0.192$
2013–2019: 1.011 (0.991–1.032) — includes 1

1.073

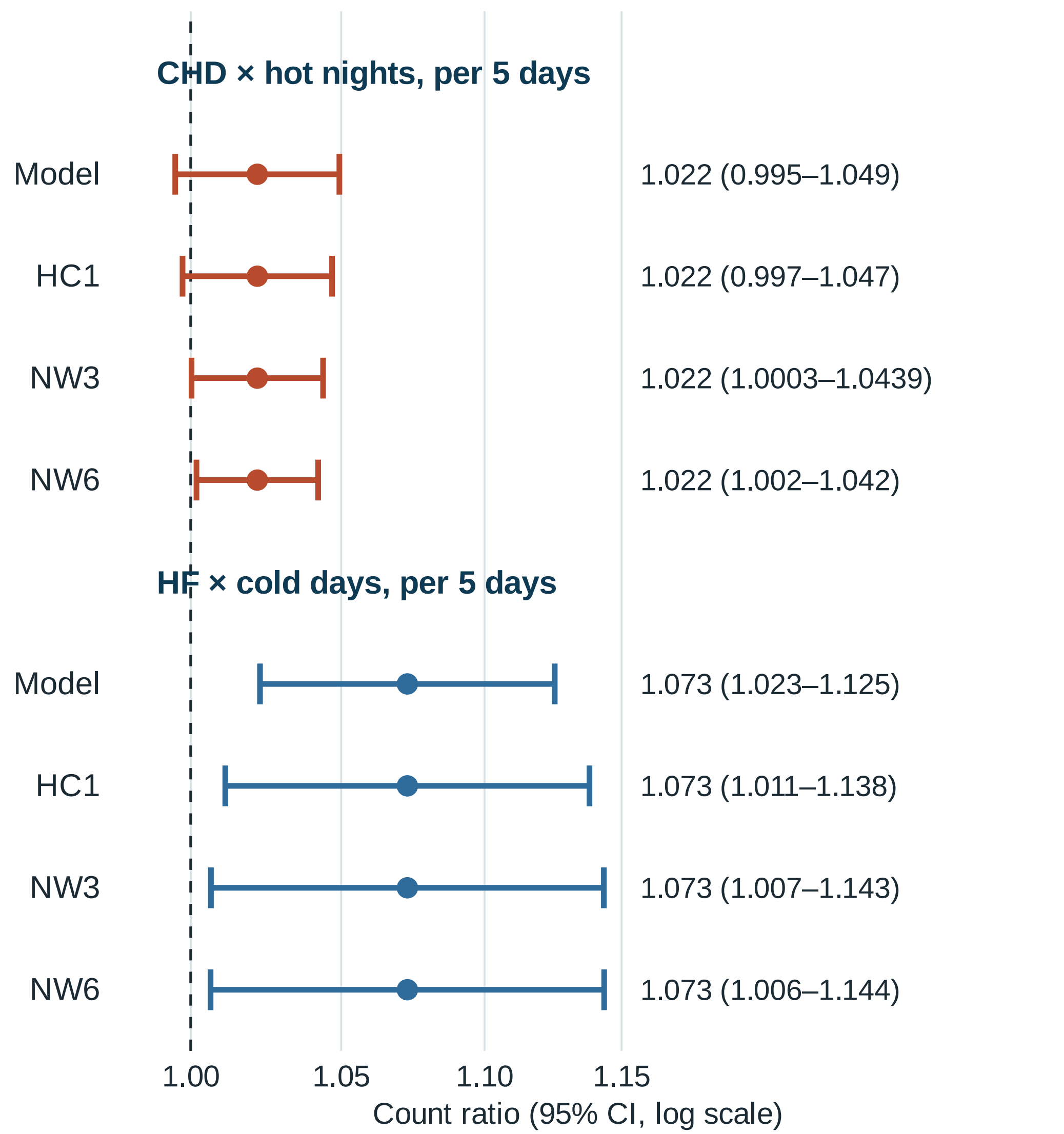
HF series × cold days, per 5 days
Full window (NW6): 1.006–1.144 $p = 0.031$ $q = 0.192$
2013–2019: 1.113 (1.053–1.176) — excludes 1

- **Unequal coherence:** the HF cold-day interval excludes 1 under all four standard-error methods and before 2020; the CHD hot-night interval includes 1 under two methods and in 2013–2019.
- The other ten contrasts sit near 1. **All 12 Benjamini–Hochberg q -values exceed 0.19:** after correction for 12 comparisons, both signals remain exploratory.



The twelve Model 1 thermal fits: count ratios from single-exposure negative-binomial models (days-in-month offset); 95% intervals with Newey–West lag-6 standard errors. Values right of 1 mean more first hospitalisations following recorded diagnosis in months with higher exposure. All 12 Benjamini–Hochberg $q > 0.19$. Models 2 and 3 are specified, not shown.

How robust are the two signals?



The two leading contrasts under four standard-error methods. For CHD × hot nights: Model and HC1 intervals include 1; NW3 excludes it only on the unrounded scale (1.0003–1.0439); NW6 excludes it. The HF cold-day interval excludes 1 under all four. The main display uses NW6; methods were not chosen by interval position.

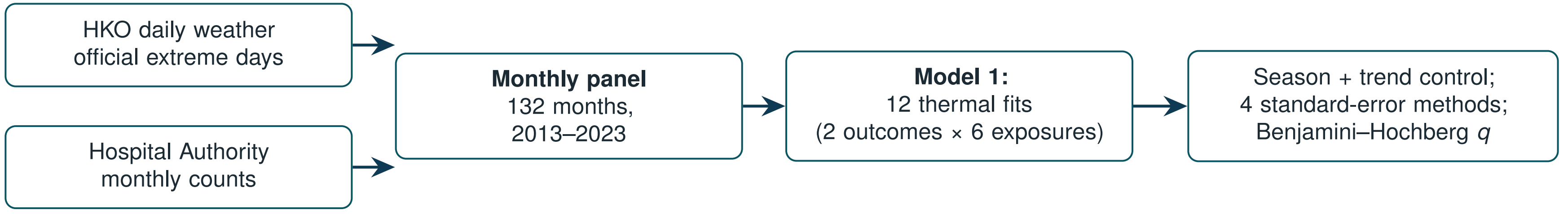
INTERPRETATION / LIMITATIONS

- All 12 q -values exceed 0.19 — these are **exploratory signals**, not confirmed effects.
- The CHD hot-night estimate is **window-dependent and uncertainty-sensitive**; the HF cold-day estimate is more coherent across methods and before 2020, but not confirmed.
- Monthly, territory-level data with no recorded admission cause support **no causal or principal-diagnosis claim**^[6].

OBJECTIVE

Are monthly temperature and extreme-day burdens associated with **first hospitalisation after a recorded coronary heart disease (CHD) or heart failure (HF) diagnosis** in a high-risk Hong Kong cohort?

METHODS



- **Window and weather:** 132 territory-months, Jan 2013 – Dec 2023; daily records and official extreme-day definitions from the Hong Kong Observatory.
- **Outcomes:** Hospital Authority monthly counts of *first hospitalisation after a first recorded CHD or HF diagnosis* in people with type 2 diabetes and/or hypertension — 156,156 CHD and 29,681 HF events. Delivered columns are labelled inpatient; admission cause is not recorded, and final event semantics require data-provider confirmation.
- **Model 1 (shown):** 2 outcomes × 6 thermal exposures = 12 negative-binomial regressions with calendar-month indicators, a trend spline, and a days-in-month offset, specified after the outcome series was available. **Model 2** adds monthly humidity and rainfall; **Model 3** counts days above or below the historical same-calendar-day mean. Models 2 and 3 are specified, not shown.
- **Uncertainty:** four standard-error methods per contrast; Benjamini–Hochberg q across all 12 comparisons.

Definitions. Hot night: minimum 28 °C or higher; very hot day: maximum 33 °C or higher; cold day: minimum 12 °C or lower (HKO definitions)^[1]. Estimates are count ratios per 5 extreme days or per 1 °C.

CONCLUSION

Monthly data suggest **different thermal patterns** in the CHD and HF series — hot nights for CHD, cold days for HF — but the evidence is **exploratory, not confirmatory**: no Model 1 contrast survived correction for twelve comparisons. Models 2–3 are specified, not shown. The project leaves a **reproducible monthly framework** and a clear next requirement: **daily or cohort data with recorded admission cause**.

What would strengthen this: linked daily weather and admission records with cause of admission, so the monthly burden patterns seen here can be tested at the resolution at which heat and cold act.

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